



LOST BOYS CYBER

THREAT INTELLIGENCE • MALWARE ANALYSIS • DETECTION ENGINEERING



Ransomware gang abuses Microsoft Teams relays to hide malicious traffic

Malware Analysis

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TLP:CLEAR | Lost Boys Cyber | Ransomware gang abuses Microsoft Teams relays to hide malicious traffic - Malware Analysis

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Ransomware gang abuses Microsoft Teams relays to hide malicious traffic

1. Executive Summary

DragonForce operators used a custom Go-based remote access backdoor, tracked by Symantec and Carbon Black as Backdoor.Turn, during an intrusion against a major U.S. services company. The most important defensive finding is the command-and-control design: Backdoor.Turn obtains an anonymous Microsoft Teams/Skype visitor token, uses Microsoft Teams-associated TURN relay infrastructure during connection setup, and then establishes a QUIC session toward attacker-controlled command infrastructure. To ordinary perimeter controls, part of the traffic profile can appear as outbound connectivity to legitimate Microsoft collaboration infrastructure.

The evidence supports a high-confidence assessment that this was a broader intrusion, not an isolated malware execution event. Reporting describes one to two months of dwell time, likely initial access through an SQL or MSSQL exposure or purchased access, downloaded tooling, DLL sideloading through VirtualBox/DbgView components, Active Directory reconnaissance, credential-enabled lateral movement, driver-assisted security tool termination, data exfiltration, and DragonForce ransomware deployment. Backdoor.Turn was reportedly installed after ransomware deployment, suggesting possible post-encryption persistence, future re-entry, or access resale.

This report is source-backed rather than sample-backed: Lost Boys Cyber did not receive malware binaries for independent detonation or reverse engineering. Static, code, and dynamic sections therefore separate reported evidence from local validation gaps. The practical defender priority is to treat this cluster as an intrusion-tradecraft alert: review SQL/MSSQL internet exposure, hunt for reported file hashes and suspicious DbgView/VirtualBox DLL sideloading, baseline outbound Teams/TURN/QUIC behavior, monitor for vulnerable driver abuse, and scope identity movement rather than limiting response to ransomware recovery.

Finding	Defensive Implication	Confidence
Backdoor.Turn abuses Teams/Skype identity and TURN relay workflow for covert C2 setup	Network allowlists that trust Microsoft collaboration endpoints without behavior analytics may miss malicious relay-assisted sessions	High for reported campaign; Medium for environment-specific detection
DragonForce combined ransomware objectives with custom tooling and BYOVD tradecraft	Treat DragonForce incidents as hands-on-keyboard intrusions with credential, driver, and lateral-movement risk	High
Backdoor.Turn reportedly arrives after ransomware deployment	Recovered environments may remain exposed if post-encryption persistence and access resale are not investigated	Medium-High
IOC set includes multiple tool, driver, DLL, and archive hashes plus staging domains	Hash-based blocking is useful but insufficient; behavior analytics are required for durable coverage	High

2. Sample Metadata

Field	Value
Family	Backdoor.Turn; DragonForce ransomware intrusion cluster
Malware Type	Go-based remote access backdoor; ransomware payload; downloader/sideloaded DLLs; vulnerable and malicious drivers
Platform	Windows enterprise endpoints and servers; Active Directory environments
Primary Sample SHA256	821da79d727351dd67ce5df7950e9a3de6647a3cf474bb3a093f6750

	7fed92a6 (reported Backdoor.Turn)
Additional Hashes	See Section 11 and `iocs.csv` for downloader, Backdoor.Turn, shellcode, DLL, vulnerable driver, AV-killer, Netscan, ADExplore, and DragonForce payload hashes
File Type / Format	Reported Go backdoor, PE DLL sideload artefacts, ZIP archives, signed vulnerable drivers, custom malicious driver, ransomware payload
Delivery Theme	Post-compromise tool deployment after likely SQL/MSSQL server exploitation or purchased access; no phishing lure confirmed in source evidence
First Seen / Reported	Activity began December 2025; public reporting published 2026-06-16
Analyst Confidence	Medium-High overall; High for source-reported observations, Medium where local sample validation is unavailable
Primary Objective	Covert remote access, reconnaissance, defense evasion, lateral movement, data theft, ransomware deployment, and possible post-ransomware persistence/access resale

3. Source Review & Confidence

Source	Contribution	Confidence
Symantec / Carbon Black Threat Hunter Team via SECURITY.COM	Primary technical account of Backdoor.Turn, Teams relay abuse, BYOVD activity, capabilities, timeline, and IOCs	High
BleepingComputer intake article	Secondary reporting that surfaced the campaign and summarized DragonForce abuse of Microsoft Teams relay infrastructure	Medium-High
Praetorian Ghost Calls research	Conceptual background for abusing web-conferencing/TURN relay flows for covert C2; not evidence of this intrusion by itself	Medium
Huntress vulnerable-driver research	Context for Huawei HWAuidoOs2Ec.sys vulnerable-driver status after the observed intrusion	Medium
Lost Boys Cyber local validation	Repository review, IOC normalization, detection/hunt engineering; no binary detonation performed for this report	Medium

Directly observed by public primary reporting: Backdoor.Turn's Teams/Skype visitor-token workflow, relay-assisted setup, QUIC C2, Go implementation, injection into DbgView64.exe, post-ransomware installation timing, DLL sideloading, vulnerable drivers, and the IOC set listed in Section 11. Assessed with high confidence: the activity represents a hands-on-keyboard ransomware intrusion with stealthy post-compromise access mechanisms. Lower-confidence or open items: the exact initial access vulnerability, the attacker's final intent for Backdoor.Turn after encryption, and the full victimology beyond the reported U.S. services-company case.

4. Initial Triage and Delivery Assessment

The campaign should be triaged as a multi-stage ransomware intrusion rather than a single malware sample. Reporting indicates the actors likely gained access through an SQL or MSSQL server vulnerability, although the specific flaw is unknown. Alternative access-broker purchase remains plausible. After initial access, the operators downloaded a ZIP archive containing legitimate executables and malicious DLL content, used DLL sideloading for execution, staged reconnaissance and evasion tooling, deployed DragonForce ransomware, and installed Backdoor.Turn afterward.

Variant / Artefact	Delivery Role	Notes
ZIP archives `9335f61...` and `cd078957...`	Tool staging package	Public reporting associates the archives with malicious deployment activity
Legitimate VirtualBox / DbgView executable	Trusted loader context	Used to load malicious DLL content and blend execution into signed or recognizable utilities
`vboxrt.dll` / sideloaded DLLs	Execution and downloader stage	Reported to download additional code for access, reconnaissance, and evasion
Backdoor.Turn	Covert remote-access stage	Go-based RAT injected into DbgView64.exe and using Teams/TURN-assisted C2 setup
Vulnerable drivers and Abyss Worker driver	Defense evasion	Used to terminate or impair security processes at kernel level
DragonForce payload	Impact stage	Data exfiltration and encryption of victim machines

Priority triage should start with systems that show SQL/MSSQL exposure, recent suspicious ZIP extraction, DbgView64.exe execution outside administrative troubleshooting windows, VirtualBox DLL side-loading from unusual directories, vulnerable driver loading, and outbound QUIC/TURN-like traffic correlated with suspicious process lineage.

5. Static Analysis

5.1. File and Build Characteristics

No local binary sample was available for independent hash verification, unpacking, or PE inspection. The following static characteristics are taken from primary reporting and normalized for defender use.

Static Signal	Evidence	Analyst Value
Go-based backdoor	Symantec tracks Backdoor.Turn as a new Go-based remote access backdoor	Go binaries often carry distinctive build artefacts, larger binary size, and recognizable runtime strings that can aid YARA development
DLL sideloading through recognizable utilities	Malicious `vboxrt.dll` and legitimate VirtualBox/DbgView execution context are reported	Supports host detections around DLL search-order abuse and signed-binary trust bypass
Signed vulnerable drivers	Reported driver set includes Topaz Antifraud, Tower of Fantasy, K7 Security Anti-Malware, and Huawei-related artefacts	Shifts detection from unsigned-driver blocking to vulnerable-driver inventory and load monitoring
Custom malicious driver	Abyss Worker driver masquerades as a Palo Alto driver	Indicates attacker willingness to deploy custom kernel-adjacent tooling rather than

		relying only on public BYOVD kits
QUIC/TURN C2 pattern	Backdoor.Turn reportedly establishes relay-assisted setup and QUIC session	Network analytics must track process, timing, destination reputation, and anomalous collaboration-service patterns

5.2. Strings, Imports, and Embedded Artefacts

Source reporting does not publish a strings listing, import table, compile timestamp, section map, or config extractor output. Defensive static review should prioritize the following if samples are acquired:

- Go build metadata, module paths, panic strings, and embedded package names that reference networking, QUIC, LDAP, browser credential stores, or process execution.
- Literal strings or encoded material related to Microsoft Teams, Skype identity services, visitor tokens, TURN relay hostnames, QUIC session setup, or relay credentials.
- DLL export names and import resolution behavior in reported VirtualBox/DbgView sideloaded DLLs.
- Driver file metadata, signer details, certificate chains, original filenames, and service-install artefacts for BYOVD components.
- Archive contents and filenames associated with `TechSupV18Fix3.zip` and the reported malicious ZIP hashes.

5.3. Static Configuration Clues

The most important reported configuration clue is Backdoor.Turn's use of Microsoft Teams/Skype-backed anonymous visitor authentication and TURN relay infrastructure to mask C2 setup. Public reporting also lists the Backdoor.Turn C2 IP `62.164.177[.J25]`, the malicious download URL `http://192.36.27[.J51/TechSupV18Fix3.zip]`, and several compromised or attacker-used domains for downloading additional tools. These indicators should be treated as time-sensitive and potentially rotated.

6. Code Analysis

6.1. Reverse-Engineering Method

Lost Boys Cyber did not perform local reverse engineering because the malware samples were not available in the task bundle. This report therefore relies on source-described behavior and provides a validation plan for future sample acquisition. Recommended follow-up tooling includes `capa`, `strings`, `FLOSS`, `radare2` or Ghidra for PE/code review, GoReSym or equivalent Go metadata parsers for Backdoor.Turn, `sigcheck`/Authenticode review for drivers, and offline config extraction focused on Teams/TURN and QUIC artefacts.

6.2. Functions and Logic of Interest

Function / Routine	Evidence	Why It Matters
Anonymous Teams visitor-token acquisition	Primary reporting states the backdoor obtains a visitor token from Microsoft's Skype-backed identity services	Explains why ordinary network monitoring may see legitimate Microsoft-associated traffic during setup
TURN relay-assisted connection setup	Backdoor.Turn uses a legitimate Microsoft TURN relay during setup	Creates a trusted-infrastructure masking problem and complicates domain/IP allowlist strategies
QUIC session to attacker C2	Reporting states a direct QUIC session is established to the real C2 server after relay-assisted setup	QUIC can reduce visibility for middleboxes that lack process-aware telemetry
Command execution and process creation	Capability list includes command execution and process creation	Enables hands-on-keyboard control, tool deployment, and follow-on lateral movement
Network scanning and service	Capability list includes TLS certificate and	Supports internal reconnaissance and target

fingerprinting	web-title capture	prioritization
LDAP / Active Directory search	Capability list includes comprehensive domain mapping	Increases identity and lateral-movement risk after endpoint compromise
Credential-based lateral movement	Capability list includes credential-based movement	Incident response must scope credential theft and remote logon activity
Browser credential theft	Capability list includes browser credential theft	Browser stores and SSO tokens become investigation priorities
Driver-assisted security termination	Reported BYOVD and Abyss Worker usage	Can disable EDR/AV and blind telemetry before ransomware impact

6.3. Anti-Analysis, Evasion, and Obfuscation

The campaign's strongest evasion layer is operational rather than traditional packer-only concealment. It combines DLL sideloading through trusted executables, legitimate Microsoft collaboration relay infrastructure, QUIC transport, signed vulnerable drivers, and custom driver tooling. Backdoor.Turn's reported injection into `DbgView64.exe` is also a process-context evasion technique, especially in environments where Sysinternals/debugging tools are occasionally used by administrators.

7. Dynamic Analysis

7.1. Execution Overview

No local sandbox run was performed. The source-reported execution chain indicates that malicious code was downloaded and executed after the operators obtained access, with subsequent reconnaissance, defense evasion, ransomware deployment, and post-impact Backdoor.Turn installation.

7.2. Filesystem, Registry, Service, and Task Changes

Reported system changes include password and access-control weakening, user/group additions, firewall rule modification, DLL placement for sideloading, driver staging/loading, and ransomware execution. Of particular concern is `LimitBlankPassword` modification, which can enable easier access to accounts with blank passwords and should be treated as both an intrusion indicator and a hardening failure.

Change Type	Reported Artefact	Defensive Interpretation
Account / group modification	New or altered users and groups	Alternative access path and potential persistence
Local security policy / registry	`LimitBlankPassword` setting changed	Weakens local access controls and aids remote access in poorly configured environments
Firewall modification	Rules changed to facilitate remote access / C2	Indicates persistence and command-channel maintenance
DLL placement	Malicious DLL mimicking VirtualBox component	Sideloading execution path
Driver load	Vulnerable or malicious drivers	Security-process termination and telemetry disruption

7.3. Network and Command-and-Control Behaviour

Backdoor.Turn's reported network behavior is unusual for ransomware operations: it obtains a Teams visitor token, interacts with Microsoft Teams/Skype-associated infrastructure, uses a TURN relay during setup, and then establishes a QUIC session

to attacker C2. This means that domain-only and IP-only blocking strategies may miss the behavior if the initial visible flow points to trusted Microsoft infrastructure. The network strategy should correlate endpoint process lineage, token/relay timing, QUIC usage, and known bad infrastructure.

7.4. Observed Capability Summary

Capability	Supporting Dynamic Evidence
Covert C2 setup	Source-reported Teams/Skype visitor-token and TURN relay workflow
Remote command execution	Backdoor.Turn capability list includes command execution and process creation
Internal reconnaissance	Network scanning, TLS certificate capture, web-title capture, LDAP/AD searches
Credential theft and lateral movement	Browser credential theft and credential-based lateral movement are reported capabilities
Defense evasion	BYOVD exploitation and Abyss Worker driver use to terminate security processes
Impact	DragonForce ransomware used for data exfiltration and encryption

8. Process Tree and Execution Chain

The public source does not provide a complete EDR process tree. The following execution chain is reconstructed from reported artefacts and should be validated against endpoint telemetry during incident response.

Step	Parent Process	Child / Action	Why It Matters
1	External-facing SQL/MSSQL service or brokered remote access	Initial operator access	Exact vulnerability unknown; scoping must include server exposure and credential reuse
2	Operator shell / remote session	Download malicious ZIP archive	Establishes staging point for tools and DLL sideload components
3	Legitimate VirtualBox or DbgView executable	Loads malicious DLL such as `vboxrt.dll`	Signed or familiar executable context reduces detection likelihood
4	Malicious DLL / loader	Downloads additional code from listed domains	Enables modular access, reconnaissance, and evasion tooling
5	Driver loader / service-control activity	Loads vulnerable drivers or Abyss Worker	Can terminate EDR/AV and reduce host visibility
6	Reconnaissance tools	ADExplore, Netscan, LDAP queries, web-title/TLS certificate capture	Maps identity, hosts, services, and lateral-movement targets
7	Ransomware deployment process	DragonForce payload executes	Data theft and encryption impact
8	DbgView64.exe context	Backdoor.Turn	Potential persistence, re-entry,

		injected/installed post-ransomware	or access resale after impact
9	Backdoor.Turn process context	Teams/Skype token request, TURN setup, QUIC C2 to `62.164.177[.]25`	Hides C2 setup among legitimate Microsoft collaboration traffic patterns

9. Persistence and System Modification

Modification Type	Artefact	Evidence	Confidence
Local access weakening	`LimitBlankPassword` configuration	Primary source reports this change for easier access	Medium-High
Account persistence	User/group additions	Primary source reports user/group changes	Medium-High
Network access persistence	Firewall rule modification	Primary source reports firewall changes to facilitate remote access and C2	Medium-High
Process-context persistence	Backdoor.Turn injected into `DbgView64.exe`	Primary source reports injection into legitimate process	High
Post-impact access	Backdoor.Turn installed after ransomware deployment	Source assesses possible persistence, future intrusion, or resale value	Medium
Security-control impairment	BYOVD and Abyss Worker driver activity	Primary source reports kernel-level process termination	High

Responders should not assume that encryption completion equals attacker departure. Backdoor.Turn's reported timing after ransomware deployment requires recovery teams to search for residual access mechanisms before restoration and reconnecting systems.

10. Network and Infrastructure Analysis

IOC / Infrastructure	Type	Role	Stability / Confidence
Microsoft Teams / Skype identity services	Legitimate cloud service	Visitor-token acquisition and relay-associated workflow	Stable service; malicious use is behavioral, not inherently bad
Microsoft Teams TURN relays	Legitimate cloud relay infrastructure	Relay-assisted C2 setup	Stable service; blocking may disrupt business; detect anomalous process use
QUIC session	Protocol / transport	Post-setup C2 communication to attacker server	Medium stability; requires endpoint and flow correlation
`62.164.177[.]25`	IP address	Reported Backdoor.Turn C2	High confidence, may rotate
`192.36.27[.]51/TechSupV18Fix3.zip`	URL / IP-hosted file	Malicious ZIP archive download URL	High confidence, likely time-sensitive
`projetosmecanicos.com[.]br` and peer domains	Domains	Download additional tools	High confidence, likely compromised or disposable

The key infrastructure lesson is that trusted-service abuse is a visibility problem, not only an IOC problem. Blocking reported domains and IPs should be immediate, but durable detection requires joining endpoint process lineage, DNS, proxy, firewall, QUIC, and identity events.

11. Indicators of Compromise

Machine-readable indicators are provided in `iocs.csv`. Bracketed domains/IPs below are defanged for safety.

IOC Type	Example / Count	Source	Operational Notes
SHA256	22 file hashes	Symantec / Carbon Black via SECURITY.COM	Includes downloader, Backdoor.Turn, shellcode, sideloaded DLLs, drivers, archives, ADExplore, AV killers, Netscan, and DragonForce payload
Domains	8 domains	Symantec / Carbon Black via SECURITY.COM	Reported C2/download infrastructure for additional tools; validate before blocking if business ownership is unclear
IP / URL	1 malicious ZIP URL, 1 Backdoor.Turn C2 IP	Symantec / Carbon Black via SECURITY.COM	Prioritize historical proxy/firewall lookups and endpoint correlation
Registry / Settings	`LimitBlankPassword`	Symantec / Carbon Black via SECURITY.COM	Configuration weakening; investigate with account and remote-logon telemetry
Process / File names	`DbgView64.exe`, `vboxrt.dll`, `HWAuidoOs2Ec.sys`, `Gamedriverx64.sys`, `wsftprm.sys`, `K7RKScan.sys`	Symantec / Carbon Black via SECURITY.COM	Names alone are not sufficient; correlate path, signer, parent process, hash, and load timing

12. MITRE ATT&CK Mapping

Technique	Name	Evidence
T1190	Exploit Public-Facing Application	Initial access reportedly likely via SQL/MSSQL server vulnerability
T1105	Ingress Tool Transfer	Operators downloaded ZIP archive and additional tools from external infrastructure
T1574.001	Hijack Execution Flow: DLL Search Order Hijacking	Malicious DLL sideloading through VirtualBox/DbgView-related execution context
T1055	Process Injection	Backdoor.Turn injected into legitimate `DbgView64.exe` process
T1090	Proxy	TURN relay-assisted command-channel setup via legitimate infrastructure
T1572	Protocol Tunneling	Backdoor.Turn uses relay-assisted setup and QUIC session to attacker C2

T1046	Network Service Discovery	Backdoor capability includes network scanning and web/TLS fingerprinting
T1087 / T1018	Account and Remote System Discovery	LDAP/AD search and domain mapping capabilities reported
T1555.003	Credentials from Web Browsers	Browser credential theft from compromised endpoints is reported
T1021	Remote Services	Credential-based lateral movement within the network is reported
T1562.001	Impair Defenses: Disable or Modify Tools	BYOVD and Abyss Worker drivers used to terminate security processes
T1543.003	Create or Modify System Process: Windows Service	Driver/service-loading activity is likely during BYOVD execution; validate per host telemetry
T1486	Data Encrypted for Impact	DragonForce ransomware encrypted victim machines
T1041	Exfiltration Over C2 Channel	Source reports data exfiltration before encryption; channel details require local validation

13. Detection Engineering

13.1. Detection Logic Summary

Signal	Telemetry Source	Analytic Goal	Tuning Notes
DbgView64.exe or VirtualBox executable loading unexpected DLLs from writable/non-install paths	EDR process/module events, Sysmon Event ID 7	Detect sideload execution path	Allow known admin/debugging directories; alert on temp, downloads, user profile, or staging paths
Driver load for reported hashes/names or vulnerable-driver families	EDR driver-load events, Sysmon Event ID 6, Code Integrity logs	Detect BYOVD / security termination preparation	Tune against approved gaming/anti-fraud software; prioritize servers and admin workstations
Security tool termination following driver load	Process events, service-control events, EDR tamper alerts	Detect defense evasion sequence	Join within 15 minutes of vulnerable driver load
Non-browser/non-Teams process initiating Teams/Skype token, TURN, or QUIC traffic	Proxy, firewall, ETW, EDR network events	Detect Backdoor.Turn-like C2 setup	Requires environment baseline for legitimate Teams components
`LimitBlankPassword` and firewall rule changes during intrusion window	Registry, Windows Firewall, audit policy events	Detect persistence/access weakening	Exclude known hardening baselines; alert on production servers
Reported IOC hits	EDR, SIEM, DNS, proxy, firewall	Identify known campaign artefacts	Treat hash hits as high-priority; infrastructure hits require process correlation

13.2. Detection Content

```

title: Suspicious DbgView Or VirtualBox DLL Sideloading In DragonForce Backdoor.Turn Intrusion
id: 9d5d7fb2-65f8-4fd2-87da-4f4e9e8c7ef1
status: experimental
description: Detects DbgView or VirtualBox-family processes loading suspicious DLLs from non-standard
locations, consistent with reported Backdoor.Turn intrusion tradecraft.
author: Lost Boys Cyber
references:
  - https://www.security.com/threat-intelligence/dragonforce-msteams-backdoor
  - https://www.bleepingcomputer.com/news/security/ransomware-gang-abuses-microsoft-teams-relays-to-
hide-malicious-traffic/
logsource:
  product: windows
  category: image_load
detection:
  selection_proc:
    Image|endswith:
      - '\DbgView64.exe'
      - '\DbgView.exe'
      - '\VirtualBox.exe'
      - '\VBoxSVC.exe'
  selection_dll:
    ImageLoaded|endswith:
      - '\vboxrt.dll'
      - '\VBoxRT.dll'
  suspicious_path:
    ImageLoaded|contains:
      - '\Users\'
      - '\ProgramData\'
      - '\Temp\'
      - '\Downloads\'
      - '\AppData\'
    condition: selection_proc and selection_dll and suspicious_path
falsepositives:
  - Portable troubleshooting utilities or non-standard VirtualBox installations; verify signer, hash,
and directory ownership.
level: high
tags:
  - attack.t1574.001
  - attack.t1055

```

```

title: Vulnerable Driver Load Associated With DragonForce BYOVD Tradecraft
id: b05a9fa6-3a71-4f70-9a82-4b4d5dc13c77
status: experimental
description: Detects loading of driver filenames reported in the DragonForce Backdoor.Turn campaign or
adjacent vulnerable-driver abuse.
author: Lost Boys Cyber
references:
  - https://www.security.com/threat-intelligence/dragonforce-msteams-backdoor
logsource:
  product: windows
  category: driver_load
detection:
  selection_name:
    ImageLoaded|endswith:
      - '\HWAuido0s2Ec.sys'
      - '\Gamedriverx64.sys'
      - '\wsftprm.sys'
      - '\K7RKScan.sys'
  selection_hash:
    Hashes|contains:
      - 'b6628d201c2a68d2a3de2a87de7a5acfe21b101a97928e1c8d5c82102d967383'
      - '252a8bb2eb9c96c5e6cc7cab822e2ed0d508032f9350351221781684e86c03ab'
      - '087f002df0a02c8c74f3ba5cd99cf29fb9efff38bf57b3d808e34a5dd4200dd2'
      - 'b16e217cdca19e00c1b68bdfb28ead53b20adeabd6edcd91542f9fbf48942877'
    condition: selection_name or selection_hash

```

```
falsepositives:
- Legitimate gaming, anti-fraud, or vendor driver installations; investigate on servers and systems without expected software.
level: high
tags:
- attack.t1562.001
- attack.t1068
```

```
rule LBC_Backdoor_Turn_Report-Based_Go_Teams_Turn_Heuristic
{
  meta:
    description = "Report-based heuristic for Backdoor.Turn-like Go binaries referencing Teams/Skype/TURN/QUIC behavior"
    author = "Lost Boys Cyber"
    reference = "https://www.security.com/threat-intelligence/dragonforce-msteams-backdoor"
    date = "2026-06-16"
    confidence = "medium - heuristic; requires sample validation"
  strings:
    $go1 = "Go build" ascii wide nocase
    $go2 = "runtime.g" ascii
    $teams1 = "teams" ascii wide nocase
    $teams2 = "skype" ascii wide nocase
    $turn1 = "turn" ascii wide nocase
    $quic1 = "quic" ascii wide nocase
    $ldap1 = "ldap" ascii wide nocase
  condition:
    uint16(0) == 0x5a4d and 1 of ($go*) and 2 of ($teams*, $turn1, $quic1, $ldap1)
}
```

```
// Microsoft Defender XDR: campaign IOC hash sweep
let CampaignHashes = dynamic([
"82b37a92589dfd4d67ca87eb9e52ac8e682e8e60d2211f59074cd5ccc693013b",
"821da79d727351dd67ce5df7950e9a3de6647a3cf474bb3a093f67507fed92a6",
"e45b18c93d187aac5c4486f57483bc87580e15def82a312bfb377ff16eb96b22",
"048e18416177de2ead251abdf4d89837f6807c6aba4d5b1debe49adfddecfb05c",
"8a4033425d36cd99fe23e6faef9764fbf555f362ebdb5b72379342fbbbe4c5531"
]);
DeviceFileEvents
| where Timestamp > ago(90d)
| where SHA256 in~ (CampaignHashes)
| project Timestamp, DeviceName, ActionType, FileName, FolderPath, SHA256, InitiatingProcessFileName, InitiatingProcessCommandLine
| union (
  DeviceProcessEvents
  | where Timestamp > ago(90d)
  | where SHA256 in~ (CampaignHashes)
  | project Timestamp, DeviceName, ActionType, FileName, FolderPath, SHA256, InitiatingProcessFileName, InitiatingProcessCommandLine
)
| order by Timestamp desc
```

13.3. Detection Validation and Blind Spots

The Sigma and YARA content above is designed from public reporting and must be validated in the client telemetry stack before enforcement. Primary blind spots include encrypted QUIC payloads, legitimate Microsoft Teams traffic volume, environments where Sysinternals and VirtualBox tools are common, and legitimate loads of named vulnerable drivers. Reduce false positives by joining multiple signals: unusual parent process, non-standard path, campaign hash, driver load, security-process termination, and suspicious outbound network timing.

14. Threat Hunting

14.1. Hunt Hypotheses

- A host compromised in this campaign may show DbgView or VirtualBox-family execution from unusual directories followed by DLL image loads, external downloads, AD reconnaissance, and driver loading.

- Backdoor.Turn-like activity may appear as non-Teams processes contacting Microsoft Teams/Skype-associated services or using QUIC/TURN patterns shortly before or after suspicious process injection.
- DragonForce operators may weaken local access controls and firewall rules before ransomware deployment.
- Driver-based security termination may precede ransomware execution and create telemetry gaps.

14.2. Hunt Queries

```
// Hunt 1: DbgView / VirtualBox sideload pattern from writable paths
let SuspiciousRoots = dynamic(["\\Users\\", "\\ProgramData\\", "\\Temp\\", "\\Downloads\\", "\\AppData\\"]);
DeviceImageLoadEvents
| where Timestamp > ago(90d)
| where InitiatingProcessFileName in~ ("DbgView64.exe", "DbgView.exe", "VirtualBox.exe", "VBoxSVC.exe")
| where FileName in~ ("vboxrt.dll", "VBoxRT.dll") or FolderPath has_any (SuspiciousRoots)
| project Timestamp, DeviceName, InitiatingProcessFileName, InitiatingProcessCommandLine, FileName,
FolderPath, SHA256, InitiatingProcessFolderPath
| order by Timestamp desc

// Hunt 2: vulnerable driver loads followed by security-process termination
let DriverNames = dynamic(["HWAuido0s2Ec.sys", "Gamedriverx64.sys", "wsftprm.sys", "K7RKScan.sys"]);
let SecurityProcesses = dynamic(["MsMpEng.exe", "SenseIR.exe", "SenseNdr.exe", "CSFalconService.exe",
"cb.exe", "carbonblack.exe", "xagt.exe", "savservice.exe"]);
let driverLoads = DeviceEvents
| where Timestamp > ago(90d)
| where ActionType has_any ("DriverLoad", "LoadImage") or FileName in~ (DriverNames)
| where FileName in~ (DriverNames) or AdditionalFields has_any (DriverNames)
| project DriverTime=Timestamp, DeviceId, DeviceName, DriverFile=FileName,
DriverFields=AdditionalFields, InitiatingProcessFileName;
let terminations = DeviceProcessEvents
| where Timestamp > ago(90d)
| where FileName in~ (SecurityProcesses)
| where ProcessCommandLine has_any ("taskkill", "stop", "kill", "terminate") or ActionType has
"Terminated"
| project TerminationTime=Timestamp, DeviceId, DeviceName, FileName, ProcessCommandLine,
InitiatingProcessFileName;
driverLoads
| join kind=innerunique terminations on DeviceId
| where TerminationTime between (DriverTime .. DriverTime + 30m)
| project DriverTime, TerminationTime, DeviceName, DriverFile, FileName, ProcessCommandLine,
InitiatingProcessFileName, DriverFields
| order by DriverTime desc

// Hunt 3: non-Teams processes using Teams/Skype-associated network paths or QUIC-like outbound traffic
let CollaborationTerms = dynamic(["teams.microsoft.com", "skype", "turn", "relay"]);
DeviceNetworkEvents
| where Timestamp > ago(30d)
| where RemoteUrl has_any (CollaborationTerms) or RemotePort in (3478, 3479, 3480, 3481, 443)
| where InitiatingProcessFileName !in~ ("Teams.exe", "ms-teams.exe", "msedge.exe", "chrome.exe",
"firefox.exe", "OUTLOOK.EXE")
| where InitiatingProcessFileName in~ ("DbgView64.exe", "DbgView.exe", "VirtualBox.exe", "VBoxSVC.exe")
or RemoteUrl has_any (CollaborationTerms)
| project Timestamp, DeviceName, InitiatingProcessFileName, InitiatingProcessCommandLine, RemoteUrl,
RemoteIP, RemotePort, Protocol, ReportId
| order by Timestamp desc

// Hunt 4: LimitBlankPassword and firewall weakening during a ransomware intrusion window
let RegistrySignal = DeviceRegistryEvents
| where Timestamp > ago(90d)
| where RegistryKey has "LimitBlankPassword" or RegistryValueName =~ "LimitBlankPassword"
| project Timestamp, DeviceId, DeviceName, Signal="Registry", ActionType, RegistryKey,
RegistryValueName, RegistryValueData, InitiatingProcessFileName, InitiatingProcessCommandLine;
let FirewallSignal = DeviceProcessEvents
| where Timestamp > ago(90d)
| where ProcessCommandLine has_any ("netsh advfirewall", "New-NetFirewallRule", "Set-NetFirewallRule",
"Enable-NetFirewallRule")
| project Timestamp, DeviceId, DeviceName, Signal="Firewall", ActionType, RegistryKey="",
RegistryValueName="", RegistryValueData="", InitiatingProcessFileName, InitiatingProcessCommandLine;
RegistrySignal
```

union FirewallSignal
order by Timestamp desc
index=windows (Image="*\\DbgView64.exe" OR Image="*\\VirtualBox.exe" OR ParentImage="*\\DbgView64.exe" OR ParentImage="*\\VirtualBox.exe")
search (ImageLoaded="*\\vboxrt.dll" OR CommandLine="*vboxrt.dll*" OR TargetFilename="*vboxrt.dll*")
table _time host user Image ParentImage CommandLine ImageLoaded TargetFilename sha256

14.3. Triage Guidance for Hunt Hits

A meaningful hit should include at least two of the following: reported campaign hash, suspicious path, unusual parent/child process relationship, driver load, security-tool termination, unexpected Teams/TURN/QUIC traffic from non-Teams processes, or firewall/local-security weakening. Benign lookalikes include legitimate Sysinternals troubleshooting, VirtualBox installations, gaming/anti-fraud drivers, and normal Microsoft Teams media traffic. Escalate to incident response when a hit involves server infrastructure, identity systems, EDR tampering, reported hashes, or ransomware-adjacent file modifications.

15. Triage and Incident Response

Priority	Action	Owner	Rationale
Critical	Isolate hosts with reported hashes, suspicious driver loads, or ransomware execution while preserving memory and disk evidence	SOC / IR	Driver-based evasion and post-impact backdoor activity can continue after encryption
Critical	Scope SQL/MSSQL exposure, logs, service accounts, and recent patch state	Infrastructure / DBA	Initial access is likely tied to SQL/MSSQL exposure or access broker pathways
High	Reset credentials for affected local, domain, service, and browser-stored accounts after containment	Identity / IR	Backdoor.Turn includes browser credential theft and credential-based lateral movement
High	Review EDR health, tamper events, and security-service termination around driver loads	SOC Engineering	BYOVD may create false confidence in “no telemetry” periods
High	Search proxy/firewall/DNS logs for listed domains, `62.164.177[.]25`, and `192.36.27[.]51` URL access	Network Security	Infrastructure hits can identify staging and C2 hosts
Medium	Baseline legitimate Teams media/TURN/QUIC traffic and identify non-standard process origins	Detection Engineering	Trusted-service abuse requires behavioral baselines
Medium	Preserve suspicious ZIP archives, DLLs, drivers, prefetch, Amcache, Shimcache, SRUM, Windows event logs, and EDR raw telemetry	Forensics	Required to reconstruct process tree and tool timing

Do not restore encrypted systems directly from backup until persistence and identity scope are complete. Pay special attention to post-ransomware Backdoor.Turn installation because it may indicate planned re-entry.

16. Validation and Emulation

Validation completed for this report: source review, IOC normalization, ATT&CK mapping, detection engineering, and threat-hunting query development. Validation not completed: sandbox execution, static unpacking, code/decompiler review, memory forensics, or network PCAP analysis of Backdoor.Turn.

Safe emulation options:

- Use Atomic Red Team or controlled scripts to emulate DLL sideload detection with benign test DLLs and DbgView/VirtualBox-like process names in a lab.
- Generate benign Sysmon driver-load test events and verify BYOVD detection logic without loading known vulnerable drivers in production.
- Replay known-bad domains and hashes as synthetic SIEM events rather than contacting live infrastructure.
- Build a lab-only QUIC/TURN traffic simulation to verify that non-Teams process network analytics fire without interacting with Microsoft production services beyond normal approved test traffic.
- Validate KQL/SPL queries against historical telemetry for false-positive rate before converting them into blocking controls.

17. Recommendations

Priority	Owner	Action	Rationale
Critical	Infrastructure / DBA	Inventory internet-facing SQL/MSSQL servers, confirm patch state, restrict exposure, and enforce MFA/strong service-account controls	Reported initial access likely involved SQL/MSSQL exposure or access broker pathways
Critical	Endpoint Security	Block or alert on the provided SHA256 indicators and vulnerable-driver hashes; enable Microsoft vulnerable driver blocklist or equivalent controls	Campaign used vulnerable and malicious drivers for defense evasion
High	Detection Engineering	Deploy the provided Sigma/KQL/SPL analytics in monitor mode, tune for legitimate Sysinternals/VirtualBox/Teams usage, then promote high-confidence joins	Durable coverage requires behavior correlation beyond IOC blocking
High	Identity	Review privileged group changes, new local/domain users, blank-password policy settings, and credential theft exposure	Operators reportedly modified access paths and Backdoor.Turn can steal browser credentials
High	Network Security	Alert on listed domains/IPs and anomalous QUIC/TURN flows from non-Teams processes	Backdoor.Turn hides setup in legitimate collaboration-service workflows
Medium	SOC	Add DragonForce Backdoor.Turn runbook steps for post-ransomware persistence checks	Backdoor installation after ransomware deployment changes recovery assumptions

Medium	Threat Hunting	Run the Section 14 hunts over at least 90 days of endpoint, identity, proxy, DNS, and firewall data	Source timeline indicates one to two months of dwell time
Medium	Forensics	If samples are recovered, perform Go metadata extraction, YARA refinement, driver signer analysis, and config extraction	Converts source-backed confidence into independently validated malware-analysis evidence

18. References and Refresh Notes

- [1] Symantec / Carbon Black Threat Hunter Team, “Hidden in Teams: DragonForce Attackers Weaponize Microsoft Teams Relays to Stay Hidden,” SECURITY.COM, accessed 2026-06-16. <https://www.security.com/threat-intelligence/dragonforce-msteams-backdoor>
- [2] BleepingComputer, “Ransomware gang abuses Microsoft Teams relays to hide malicious traffic,” published 2026-06-16. <https://www.bleepingcomputer.com/news/security/ransomware-gang-abuses-microsoft-teams-relays-to-hide-malicious-traffic/>
- [3] Praetorian, “Ghost Calls: Abusing web conferencing for covert command and control,” referenced by primary reporting for technique background. <https://www.praetorian.com/blog/ghost-calls-abusing-web-conferencing-for-covert-command-control-part-1-of-2/>
- [4] Huntress, Huawei `HWAuidoOs2Ec.sys` vulnerable-driver research, referenced by primary reporting for vulnerable-driver context.
- [5] MITRE ATT&CK Enterprise Matrix, techniques cited in Section 12. <https://attack.mitre.org/>

Refresh guidance: re-check primary reporting and vendor bulletins before client delivery because C2 infrastructure, vulnerable-driver blocklist coverage, Microsoft Teams/TURN detection guidance, and DragonForce victimology may change quickly. If malware samples become available, update Sections 5-7 with real hash verification, strings/imports, code paths, dynamic execution evidence, packet captures, and a concrete process tree.